

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVC504 - CONCRETE TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the primary role of water in concrete?
 - a) To increase workability
 - b) To activate hydration of cement
 - c) To reduce the cost of construction
 - d) To act as a binder
- 2 Bulking of sand is primarily caused by:
 - a) Temperature
 - b) Silt content
 - c) Moisture content
 - d) Aggregate size
- 3 Which method is NOT used to measure the workability of concrete?
 - a) Slump test
 - b) Vee Bee test
 - c) Vicat Apparatus test
 - d) Compaction factor test
- 4 Segregation in fresh concrete is defined as:
 - a) Uniform distribution of water and aggregates
 - b) Separation of coarse aggregates from the rest of the mix
 - c) Increase in air content
 - d) Decrease in workability
- 5 Which of the following is true about high-performance concrete?
 - a) It has lower strength than conventional concrete
 - b) It is designed for better durability and workability
 - c) It is only used in low-strength applications
 - d) It requires no curing

- 6 Which admixture increases the workability of concrete without increasing water content?
- Retarder
 - Superplasticizer
 - Accelerator
 - Pozzolana
- 7 Green concrete refers to:
- High strength concrete
 - Concrete mixed with green pigment
 - Environmentally friendly concrete
 - Concrete that is not fully cured
- 8 Which of the following factors contributes to the corrosion of steel reinforcement in concrete?
- Sulphate attack
 - Chloride attack
 - Alkali-aggregate reaction
 - Thermal expansion
- 9 Characteristic strength of concrete is defined as strength below which not more than:
- 10 % of results fall
 - 5% of results fall
 - 50% of results fall
 - 25% of results fall
- 10 W/C ratio mainly affects:
- Durability
 - Workability
 - Strength
 - All of the above
- 11 Creep in concrete is:
- Sudden deformation
 - Time-dependent deformation
 - Due to poor mix
 - Permanent expansion
- 12 Non-destructive testing method used to determine concrete strength is:
- Core cutting
 - Rebound hammer test
 - Slump test
 - Compaction test

Q2. Solve the following.

- Explain the manufacturing process of Ordinary Portland Cement (OPC) with a neat flow diagram. 6
- Explain types of cement. 6

Q3. Solve the following.

- Define workability. Explain slump cone test for measuring workability 6

- B) Differentiate between volume batching and weight batching. 6
- Q4. Solve Any Two of the following.
- A) What are admixtures? Explain the classification and roles of different types of admixtures in concrete. 6
- B) Illustrate difference between accelerator and plasticizer. 6
- C) What is green concrete? Discuss its benefits and applications in sustainable construction. 6
- Q5. Solve Any Two of the following.
- A) What is Alkali Aggregate Reaction (AAR) and explain factors affecting on AAR. 6
- B) Define creep. Explain factors affecting Creep. 6
- C) Compare high-strength concrete and high-performance concrete. 6
- Q6. Solve Any Two of the following.
- A) Explain the process of concrete mix design. What are the factors influencing mix design? 6
- B) What are the commonly used non-destructive tests (NDT) for concrete? Describe any two in detail. 6
- C) Explain quality measurement in Concrete Construction. 6

*** End ***